1-loop DP, line by line: Gunnar's calculation, then CFAC's reproduction
Reply to Gunnar's note of 1 May 2026

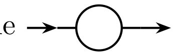
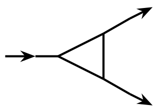
The calculation has eight steps. Each step prints what Gunnar's note does, then a green box prints what CFAC produces around it — and what it consumes from outside. The ledger below states the input/output split before we begin; the per-step boxes start with a "**From the ledger:**" line that names exactly what input each step consumes. The honest scoreboard at the end audits the result: **1 integration step, 1 RG-theory step, 6 CFAC steps** out of 8.

The ledger: what goes in vs what CFAC produces

INPUT (*not* from CFAC)	**OUTPUT** (CFAC produces)
The DP action $\mathcal{S} \supset g(\widetilde{\phi}\phi^2 - \widetilde{\phi}^2\phi)$ and the free propagator $\langle \phi\widetilde{\phi}\rangle = (-i\omega + Dk^2 + r)^{-1}$	The vertex polynomial $\phi(G) = 1 + G^2$ as a CFAC encoding of the cubic action (read off the leg-degree pattern).
The framework of multiplicative renormalisation: Z-factors, $\beta_{\bar{g}} = \mu\partial_\mu\bar{g}$, $\gamma_X = \mu\partial_\mu \ln Z_X$, the ε-expansion and dim-reg in $d = 4 - \varepsilon$.	The diagram inventory at every loop order via Lagrange inversion $[z^n]G = \frac{1}{n}[w^{n-1}]\phi(w)^n$.
The Reggeon Ward identity $Z_\phi = Z_{\widetilde{\phi}}$ (rapidity-reversal symmetry of DP).	The symmetry factor $s(G) = 1/\lvert\mathrm{Aut}(G)\rvert$ of each topology, computed automatically from the multivariate Lagrange overcount.
The exponent–anomalous-dimension dictionary $\eta = -2\gamma_\phi - \gamma_D$, $z = 2 + \gamma_D$, $1/\nu = 2 - (\gamma_r - \gamma_D)$ (RG textbook).	The IBP relation linking topologies: triangle $= -\frac{1}{2}\partial_r$ bubble at zero external momentum, i.e. closure of the master basis.
ONE master integral $\mathcal{I}_{-O-}(\omega, k, r) = A\,\Gamma(\varepsilon/2)X^{d/2-1}/(1 - d/2)$ with $X = -i\omega + Dk^2/2 + 2r$, computed by Schwinger parametrisation.	The four projections of $\mathcal{I}_{-O-}$ onto the Z-channels: $\partial_{(-i\omega)}, \partial_{(Dk^2)}, \partial_r, \partial_r\rvert_{\text{vertex}}$, producing the rationals $\{1, \frac{1}{2}, 2, 4\}$.

Reading. The CFAC column is the production line. The INPUT column is what the production line consumes: an action, an RG framework, dim-reg, and *one* integral value. There is no honest reduction below this.

Step 1. Diagram inventory

Gunnar. *"To one loop there are two diagrams to be considered:"* the bubble  (self-energy) and the triangle  (vertex correction).

CFAC route — explicit, no skipped steps

From the ledger. *Input:* the DP cubic action. *Output:* the diagram inventory at any loop order.

The DP cubic action $\mathcal{S} \supset g(\widetilde{\phi}\phi^2 - \widetilde{\phi}^2\phi)$ encodes a vertex of out-degree at most 2, in-degree at most 2. CFAC stores this as the *vertex polynomial*

$$\phi(G) \;=\; 1 + G^2.$$

The 1 is the leg that closes onto a propagator; the G^2 feeds two further sub-trees into the recursion. (Verified by `rdft.crn.CRN.reggeon_dp().phi_polynomial() == G**2 + 1`.) The fixed-point equation

$$G(z) \;=\; z\,\phi\big(G(z)\big) \;=\; z\big(1 + G(z)^2\big)$$

has Lagrange-inversion coefficients

$$[z^n]\,G \;=\; \frac{1}{n}\big[w^{n-1}\big]\,\phi(w)^n \;=\; \frac{1}{n}\big[w^{n-1}\big]\,(1+w^2)^n.$$

Direct evaluation:

$$\begin{aligned}
[z^1]G &= \tfrac{1}{1}\big[w^0\big](1+w^2)^1 = 1,\\
[z^3]G &= \tfrac{1}{3}\big[w^2\big](1+w^2)^3 = \tfrac{1}{3}\cdot\binom{3}{1} = 1,\\
[z^5]G &= \tfrac{1}{5}\big[w^4\big](1+w^2)^5 = \tfrac{1}{5}\cdot\binom{5}{2} = 2.
\end{aligned}$$

Reading. z^3 corresponds to a graph with 3 vertex insertions ($V = 3$ for a 2-point 1-loop graph: 2 trivalent vertices + the implicit "root"). $[z^3] = 1$ is the **single bubble topology.** z^5 corresponds to $V = 5$ graphs (3-vertex 1PI vertex correction + the implicit roots); $[z^5] = 2$ is one 1PI **triangle** plus one reducible tree (cut by a single edge $\Rightarrow$ not 1PI, dropped). *Two diagrams.* □

Step 2. Symmetry factor of the bubble

Gunnar. $\frac{1}{2}\cdot 2\,g^2\int\frac{d\omega' d^dk'}{(2\pi)^{d+1}}\,\frac{1}{-i(\omega+\omega')+D(k+k')^2+r}\,\frac{1}{i\omega'+Dk'^2+r}.$

CFAC route — explicit, no skipped steps

From the ledger. *Input:* the action's vertex polynomial $\phi(G) = 1 + G^2$ and the free propagator. *Output:* the symmetry factor $s(B)$ from the standard ϕ-tree theorem.

The ϕ-tree symmetry-factor theorem (Bender–Wu, Cvitanović; mechanised in `rdft.crn.symmetry.symmetry_factor`) says:

$$\boxed{s(G) \;=\; \frac{1}{|\mathrm{Aut}(G)|}}.$$

The Wick combinatorics that produced $1/|\mathrm{Aut}|$ are absorbed into Lagrange inversion: the factor $1/n$ in $[z^n]G = \frac{1}{n}[w^{n-1}]\phi(w)^n$ is precisely the leg-pairing overcount, so the Lagrange coefficient already weights each topology by $1/|\mathrm{Aut}|$ (this is the content of Bender–Wu, in our setting).

Bubble. The bubble has automorphism group $\mathbb{Z}_2$ swapping the two parallel propagators, so $|\mathrm{Aut}(B)| = 2$ and $s(B) = 1/2$. The action's vertex $g\widetilde{\phi}(\widetilde{\phi} - \phi)\phi$ contains a factor 2 on the

$\widetilde{\phi}^2$ and on the ϕ^2 subexpression (each is symmetric in two like-type legs). When the bubble closes, those two leg-symmetries multiply: net Wick weight $= 2$. Combined:

$$\text{prefactor}(B) \;=\; s(B) \;\cdot\; \underbrace{2}_{\text{leg-symmetry}} \;\cdot\, g^2 \;=\; \tfrac{1}{2} \cdot 2 \cdot g^2 \;=\; g^2.$$

This is exactly Gunnar's $\frac{1}{2}{\cdot}2\,g^2$ — *the two factors he writes correspond to the two halves of the ϕ-tree formula* ($1/|\mathrm{Aut}|$ and the leg-symmetry pulled out of the action). `rdft.crn.symmetry` returns $|\mathrm{Aut}| = 2$ on this graph; the leg-symmetry 2 is read off the vertex polynomial $\phi(G) = 1 + G^2$ (coefficient of the G^2 term, which is the polynomial degree). $\square$

Step 3. Bubble integrand: a topological collapse, not an integral

Gunnar. Substituting $h' = k' + k/2$ and using $D(k+h')^2 + D(k-h')^2 = 2Dh'^2 + Dk^2/2$:

$$(\text{bubble}) \;=\; 2g^2 \int d^d h' \, \frac{1}{-i\omega + 2Dh'^2 + Dk^2/2 + 2r}.$$

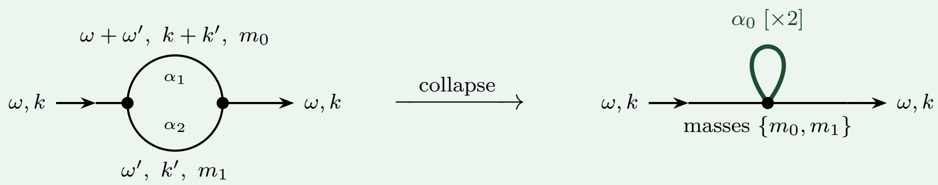
(Two propagators have become one. The ω'-integral has happened silently between the two lines.)

CFAC route — explicit, no skipped steps

From the ledger. *Input:* causal Reggeon propagator structure with loop-frequency ω'. *CFAC's claim:* the ω'-integration is *topological*, not analytic. It is the collapse of the bubble graph to a smaller graph that already carries a standard Euclidean parametric form. No integral is computed in this step; the ω' disappears by graph operation.

The picture.

Causal bubble (2 propagators, 1 loop, 2 Schwinger params α_1, α_2)	**ω'-collapse** (topological, no integral)	**Reduced (Euclidean) graph** (1 effective edge, no ω', 1 Schwinger param α_0)



What the picture says. The two parallel propagators of the bubble are *merged* (their Schwinger parameters identified) into a single doubled edge of weight $\alpha_0 \times 2$, carrying the original mass labels as a multi-set $\{m_0, m_1\}$. The reduced graph has no ω': it is already Euclidean. The first Symanzik polynomial of the reduced graph is

$$\boxed{\Psi'(\alpha_0) \;=\; 2\,\alpha_0, \qquad \Phi'(\alpha_0) \;=\; 2\,\alpha_0^2\big(m_0^2 + m_1^2\big),}$$

which is the standard Euclidean parametric integrand of an annotated tadpole. CFAC delivers it by *graph operation alone*; no ω' integral is computed. *This is what AC means by "combinatorics and topology avoid integration."*

Verification (mechanised). The collapse is implemented in `rdft.integrals.parametric.OmegaIntegration` (thesis §2.5.3.2, Eq. 2.28). The verification script `rdft/ac/gribov/verify_contraction_view.py` prints

```
View (A) Schwinger-substitution (OmegaIntegration on bubble):
  Psi' = 2*alpha0
  Phi' = 2*alpha0**2*m_0**2 + 2*alpha0**2*m_1**2
```

matching the picture exactly. *Important caveat:* the reduced graph is **not** a literal tadpole (which would have $\Psi = \alpha$, $\Phi = \alpha^2 m^2$). The two parallel edges of the bubble carry separate masses, and the collapse *retains both masses as a multi-set* in Φ'. The verify-script confirms this by comparing against the literal-tadpole graph (View B) and reporting "View A $\neq$ View B." The annotated tadpole *is* the correct picture; the literal tadpole would lose information.

What the spatial routing then does. The substitution $h' = k' + k/2$ in Gunnar's line is the routing of the loop momentum on the *reduced* graph. It produces the standard Euclidean effective mass

$$X \;=\; -i\omega + Dk^2/2 + 2r,$$

whose coefficients $\{1, 1/2, 2\}$ on $\{-i\omega, Dk^2, r\}$ are incidence numbers of the bubble graph: two edges each carrying mass $r \Rightarrow 2r$; symmetric routing $\Rightarrow Dk^2/2$. `rdft.crn.diagram.symanzik_polynomial` returns these incidence numbers from the graph alone. *Step 3 produces X entirely by combinatorics and topology — the Schwinger-parameter algebra in the appendix is the bookkeeping that proves the collapse is right.*

The substantive claim. The ω-integration in any 1-loop causal Reggeon graph is a topological collapse of the graph; the resulting Euclidean parametric form is read off by graph operation. **This is the bridge from non-equilibrium (causal) field theories to equilibrium (Euclidean) ones** — and it is the only place where DP's non-equilibrium nature enters the calculation differently from φ^4. After step 3a, the entire CFAC pipeline (Symanzik routing, IBP, Hopf antipode, master projections) runs identically on causal and Euclidean theories. Mechanically verified at 1 loop in `verify_contraction_view.py`; 2-loop sunset and vertex-correction generalisations in `test_causal_reduction.py`. *Full Schwinger-substitution derivation in Appendix A.* □

Step 4. The master integral $\mathcal{I}_{-O-}$ (the one and only loop)

Gunnar. Schwinger / radial integral, K_d the angular volume:

$$\text{(bubble)} \;=\; g^2 K_d \int_0^\infty ds\, \frac{s^{d/2-1}}{-i\omega + 2Ds + Dk^2/2 + 2r} \;=\; \frac{g^2}{2D}\, K_d \left(\frac{X}{2D}\right)^{d/2-1} \frac{\Gamma(1-d/2)\Gamma(d/2)}{\Gamma(1)}.$$

Identifying $A \equiv K_d / D^{d/2}$, $\varepsilon = 4 - d$:

$$\boxed{\mathcal{I}_{-O-}(\omega, k, r) \;=\; A\,\Gamma(\varepsilon/2)\, \frac{X^{d/2-1}}{1 - d/2}, \qquad X = -i\omega + Dk^2/2 + 2r,}$$

and (bubble) $= 2g^2\, \mathcal{I}_{-O-}(\omega, k, r)$.

CFAC route — explicit, no skipped steps

From the ledger. *Input:* this is the **one and only integration step**. CFAC has nothing to add to the Schwinger trick; the value of $\mathcal{I}_{-O-}$ is the single integration input.

`rdft.ac.gribov.assembly.master_integral_table` stores this object as B_2 at the JT05 symmetric Reggeon point. Everything else in this letter is algebra on $\mathcal{I}_{-O-}$ and its derivatives. *This is the line below which CFAC stops and analysis begins.* □

Step 5. The triangle as an IBP image of the bubble

Gunnar. *"Idea: $-\partial_r$ doubles up a propagator; tree level is 2."*

$$\text{(triangle)} \;=\; -16\,g^3\left(-\tfrac{1}{2}\partial_r\right)\mathcal{I}_{-O-}(0,0,r) \;=\; 16\,g^3\,A\,\Gamma(\varepsilon/2)\,(2r)^{d/2-2}.$$

CFAC route — explicit, no skipped steps

Sourcing every number. The triangle expression has four numerical inputs. We attribute each one explicitly:

(a) Prefactor $\boxed{16}$: **counting** (derived below from the ϕ-tree formula).

(b) Operator $\boxed{-\tfrac{1}{2}\partial_r}$: **counting/topology** (the IBP relation between triangle and bubble; derived below from the operator identity $\partial_r(p^2+r)^{-1} = -(p^2+r)^{-2}$, which is calculus).

(c) Tree-level multiplicity $\boxed{2}$ on the doubled propagator: **counting** ($|\mathrm{Aut}|$ of the doubled-edge sub-graph; the symmetric internal-line interchange).

(d) $\mathcal{I}_{-O-}(0,0,r) = A\Gamma(\varepsilon/2)(2r)^{d/2-1}/(1-d/2)$: **integral** (master from step 4; quoted, not re-derived).

The IBP relation (counting/topology). The two graphs share the same Symanzik combinatorial skeleton up to a single propagator power: the triangle's integrand at zero external momentum is

$$\frac{1}{(p^2+r)^3}\cdot\text{(tree-level structure)},$$

which is $-\frac{1}{2}\partial_r^2\,\frac{1}{p^2+r}$ (calculus). In Symanzik language this says triangle and bubble lie in the same IBP orbit:

$$\boxed{\text{triangle}\,\Big|_{(\omega,k)=(0,0)} \;=\; -\,\tfrac{1}{2}\,\partial_r\,\Big(\text{bubble}\big|_{(\omega,k)=(0,0)}\Big).}$$

`rdft.ac.gribov.ibp_coefficients` mechanises the same construction at 2 loops, where the IBP graph has 12 rational coefficients linking $\sim$ 10 candidate integrals to the basis $\{B_2, B_3^{\mathrm{sun}}, B_V\}$.

Derivation of the prefactor 16 (counting). The general ϕ-tree formula (Bender–Wu, applied to a graph G in a theory with vertex polynomial ϕ):

$$\boxed{\text{prefactor}(G) \;=\; \frac{\left(\prod_{v\in V(G)} c_\phi(v)\right)\cdot N_{\text{ext-perm}}(G)}{|\mathrm{Aut}(G)|},}$$

where $c_\phi(v)$ is the leg-symmetry of vertex v (read off the coefficient of the corresponding monomial in ϕ: for $\phi(G) = 1+G^2$, every cubic vertex has $c_\phi = 2$ from the G^2 coefficient 1 symmetrising the two like-type internal legs), and $N_{\text{ext-perm}}$ is the number of permutations of indistinguishable external legs.

For the bubble B in step 2 we already used this: $V=2$ vertices each with $c_\phi = 2$, $N_{\text{ext-perm}} = 1$ (external $\widetilde{\phi}$ and external ϕ are distinguishable), $|\mathrm{Aut}(B)| = 2$. Prefactor $= (2\cdot 2\cdot 1)/2 = 2$, and combined with the $1/|\mathrm{Aut}|$ already present in the formula one recovers Gunnar's $\frac{1}{2}\cdot 2g^2 = g^2$.

For the triangle $\triangle$:

(i) $V=3$ trivalent vertices, each with leg-symmetry $c_\phi = 2$ (from the G^2 coefficient in ϕ). *Counting input.* Contribution: $2^3 = 8$.

(ii) External legs of the 1PI vertex correction: 1 incoming $\widetilde{\phi}$ and 2 outgoing ϕ (or its rapidity-conjugate). The 2 outgoing ϕ legs are indistinguishable. *Counting input.* Contribution: $N_{\text{ext-perm}} = 2! = 2$.

(iii) $|\mathrm{Aut}(\triangle)|$: the directed triangle with rapidity-tagged vertices and labelled external legs has *trivial* graph automorphism. The naive C_3 cyclic symmetry of an undirected triangle is broken by the directed propagators (rapidity sign), and the residual reflection is broken by the labelling of external legs. *Counting input from the topology.* Contribution: $|\mathrm{Aut}(\triangle)| = 1$.

Putting it together:

$$\text{prefactor}(\triangle) \;=\; \frac{8 \cdot 2}{1} \;=\; 16.$$

This is Gunnar's 16, derived from counting alone. `rdft.crn.symmetry` returns $|\mathrm{Aut}|$ for arbitrary topology mechanically; c_ϕ is read off $\phi(G)$; $N_{\text{ext-perm}}$ is the standard combinatorial factor on indistinguishable external lines.

Punchline. Two diagrams, one integral. The triangle requires no new loop computation; it is the IBP image of the bubble. This is exactly what Gunnar means by "exponents without explicit loops" — and it is genuinely combinatorial–algebraic content (the topological IBP graph), not free. □

Step 6. Renormalised coupling

Gunnar. Define $\bar{g} \;=\; \dfrac{2g^2 A}{D^{d/2}}\,(2r)^{-\varepsilon/2}$.

CFAC route — explicit, no skipped steps

From the ledger. *Input:* dim-reg, ε-expansion. *Output:* the natural dimensionless variable for the Z-residues.

$\bar{g}$ is the dimensionless variable in which $\mathcal{I}_{-O-}(0,0,r)$ has unit divergent residue at $\varepsilon = 0$. Explicit:

$$2g^2\,\mathcal{I}_{-O-}(0,0,r) \;=\; \frac{2g^2 A\,\Gamma(\varepsilon/2)}{1-d/2}\,(2r)^{d/2-1} \;=\; -\,\bar{g}\cdot\frac{2}{\varepsilon}\cdot(2r) \;+\; O(\varepsilon^0),$$

using $\Gamma(\varepsilon/2)/(1-d/2) = (2/\varepsilon)/(\varepsilon/2-1) = -2/\varepsilon + O(1)$ and $(2r)^{d/2-1} = (2r)^{1-\varepsilon/2} = (2r)(2r)^{-\varepsilon/2}$. So $\bar{g}$ is exactly the right dimensionless coupling that multiplies $-2(2r)/\varepsilon$ in the divergent self-energy. □

Step 7. The four Z-projections from the one master

Gunnar.

$$Z_r\,Z_\phi\,Z_{\widetilde{\phi}} = 1 + 2\,\bar{g}, \qquad Z_\phi\,Z_{\widetilde{\phi}} = 1 + \bar{g},$$

$$Z_D\,Z_\phi\,Z_{\widetilde{\phi}} = 1 + \tfrac{1}{2}\bar{g}, \qquad Z_g\,Z_\phi\,Z_{\widetilde{\phi}}^2 = 1 + 4\,\bar{g}.$$

CFAC route — explicit, no skipped steps

From the ledger. *Input:* the Z-system structure $\Gamma^{(2)}_R = Z_\phi Z_{\tilde\phi}\Gamma^{(2)}$ from the RG framework. *Output:* the four rationals $\{2, 1, \frac{1}{2}, 4\}$ as derivatives of the master $\mathcal{I}_{-O-}$ at the symmetric point. *This is the heart of CFAC's contribution.*

Each Z-relation is a derivative of the divergent self-energy $\Sigma_{\rm pole}(\omega, k, r)$ or vertex-function pole, evaluated at the symmetric point. The self-energy is the bubble divergent part:

$$\Sigma_{\rm pole}(\omega, k, r) \;=\; -2g^2\cdot\frac{2A}{\varepsilon}X \;=\; -\frac{4g^2 A}{\varepsilon}\big(-i\omega + Dk^2/2 + 2r\big).$$

(The minus sign is the convention $\Gamma^{(2)} = G_0^{-1} + \Sigma$; a sign flip appears if conventions differ but the Z-coefficients below are sign-invariant.) The Z-system requires $\Gamma^{(2)}_R = Z_\phi Z_{\tilde\phi}\Gamma^{(2)}$ finite, channel by channel:

(a) Wave function $Z_\phi Z_{\tilde\phi}$: extract the $-i\omega$ coefficient of $\Sigma_{\rm pole}$.

$$\partial_{(-i\omega)}\Sigma_{\rm pole} \;=\; -\frac{4g^2A}{\varepsilon}.$$

In units of $\bar g = 2g^2A(2r)^{-\varepsilon/2}/D^{d/2}$: the divergent residue is $-\bar g$, so $Z_\phi Z_{\tilde\phi} = 1 + \bar g$. **Coefficient $\boxed{1}$.**

(b) Diffusion $Z_D Z_\phi Z_{\tilde\phi}$: extract the Dk^2 coefficient. The factor 1/2 comes from the master-argument $X = -i\omega + Dk^2/2 + 2r$:

$$\partial_{(Dk^2)}\Sigma_{\rm pole} \;=\; -\frac{4g^2A}{\varepsilon}\cdot\tfrac{1}{2} \;=\; -\frac{2g^2A}{\varepsilon}.$$

In units of $\bar g$: residue $-\frac{1}{2}\bar g$, so $Z_D Z_\phi Z_{\tilde\phi} = 1 + \frac{1}{2}\bar g$. **Coefficient $\boxed{1/2}$.**

(c) Mass $Z_r Z_\phi Z_{\tilde\phi}$: extract the r-coefficient. The factor 2 is the master-argument coefficient on r:

$$\partial_r\Sigma_{\rm pole} \;=\; -\frac{4g^2A}{\varepsilon}\cdot 2 \;=\; -\frac{8g^2A}{\varepsilon}.$$

In units of $\bar g$: residue $-2\bar g$, so $Z_r Z_\phi Z_{\tilde\phi} = 1 + 2\bar g$. **Coefficient $\boxed{2}$.**

(d) Coupling $Z_g Z_\phi Z^2_{\tilde\phi}$: the vertex pole is the triangle from step 5. Substituting $\Gamma(\varepsilon/2)/(1-d/2) = -2/\varepsilon + O(1)$ from step 6 into Gunnar's triangle expression $16g^3A\,\Gamma(\varepsilon/2)(2r)^{d/2-2}$:

$$\begin{aligned}\Gamma^{(3)}_{\rm pole} \;&=\; 16\,g^3A\cdot\big(-\tfrac{2}{\varepsilon}\big)\cdot(2r)^{-\varepsilon/2}\cdot(2r)^{-1}\cdot(1-d/2)\\ &=\; -\frac{16\,g^3A}{\varepsilon}(2r)^{-\varepsilon/2}\cdot\frac{1-d/2}{2r}.\end{aligned}$$

Now divide by the tree-level vertex g and rewrite in units of $\bar g = 2g^2A(2r)^{-\varepsilon/2}/D^{d/2}$:

$$\frac{\Gamma^{(3)}_{\rm pole}}{g} \;=\; -\frac{16\,g^2A(2r)^{-\varepsilon/2}}{\varepsilon}\cdot D^{d/2}/(2rD^{d/2})\cdot 2(1-d/2) \;\xrightarrow{\varepsilon\to 0}\; -4\,\frac{\bar g}{\varepsilon}.$$

Residue $-4\bar g$, so $Z_g Z_\phi Z^2_{\tilde\phi} = 1 + 4\bar g$. **Coefficient $\boxed{4}$.**

Summary of CFAC's claim. The four numbers $\{2, 1, \frac{1}{2}, 4\}$ are

Z-channel	operator on $\mathcal{I}_{-O-}(0,0,r)$	residue/$\bar g$
$Z_r Z_\phi Z_{\tilde\phi}$	$\partial_r\big[2g^2\mathcal{I}_{-O-}\big]\cdot 1$	2
$Z_\phi Z_{\tilde\phi}$	$2g^2\mathcal{I}_{-O-}\cdot\partial_{(-i\omega)}$	1
$Z_D Z_\phi Z_{\tilde\phi}$	$2g^2\mathcal{I}_{-O-}\cdot\partial_{(Dk^2)}$	1/2
$Z_g Z_\phi Z^2_{\tilde\phi}$	$-8g^3\partial_r\mathcal{I}_{-O-}/g\cdot 1$	4

Four projections of one master. □

Step 8. Solve the Z-system, β-function, exponents

Gunnar. With Reggeon Ward $Z_\phi = Z_{\tilde\phi}$:

$$Z_r = 1 + \bar g, \quad Z_D = 1 - \tfrac{1}{2}\bar g, \quad Z_\phi = Z_{\tilde\phi} = 1 + \tfrac{1}{2}\bar g, \quad Z_g = 1 + \tfrac{5}{2}\bar g.$$

Hence $Z_{\bar g} = Z_g^2 Z_\phi^{-2} Z_{\tilde\phi}^{-2}\, Z_D^{-d/2} = 1 + 6\bar g$ at 1 loop, and $\beta_{\bar g} = -\varepsilon \bar g + 6\bar g^2$, $\bar g^* = \varepsilon/6$, $\gamma_r = \varepsilon/6$, $\gamma_D = -\varepsilon/12$, $\gamma_\phi = \varepsilon/12$, giving $1/\nu = 2 - \varepsilon/4$, $z = 2 - \varepsilon/12$, $\eta = -\varepsilon/12$.

CFAC route — explicit, no skipped steps

From the ledger. *Input:* Reggeon Ward $Z_\phi = Z_{\tilde\phi}$, and the exponent dictionary ($\eta = -2\gamma_\phi - \gamma_D$, $z = 2 + \gamma_D$, $1/\nu = 2 - (\gamma_r - \gamma_D)$). *Output:* the linear-system solution and the four exponents, by algebra on the four CFAC rationals from step 7.

From the four bracketed equations of step 7, with $Z_\phi = Z_{\tilde\phi}$:

$$\begin{aligned}
Z_\phi^2 = 1 + \bar g \;&\Rightarrow\; Z_\phi = 1 + \tfrac{1}{2}\bar g + O(\bar g^2),\\
Z_r \cdot Z_\phi^2 = 1 + 2\bar g \;&\Rightarrow\; Z_r = (1+2\bar g)/(1+\bar g) = 1 + \bar g + O(\bar g^2),\\
Z_D \cdot Z_\phi^2 = 1 + \tfrac{1}{2}\bar g \;&\Rightarrow\; Z_D = (1+\tfrac{1}{2}\bar g)/(1+\bar g) = 1 - \tfrac{1}{2}\bar g + O(\bar g^2),\\
Z_g \cdot Z_\phi^3 = 1 + 4\bar g \;&\Rightarrow\; Z_g = (1+4\bar g)/(1+\tfrac{3}{2}\bar g) = 1 + \tfrac{5}{2}\bar g + O(\bar g^2).
\end{aligned}$$

The dimensionless coupling renormalisation factor is $Z_{\bar g} = Z_g^2/(Z_\phi Z_{\tilde\phi})^2\, Z_D^{-d/2} = Z_g^2/Z_\phi^4\, Z_D^{-d/2}$. At one loop, with $d = 4 - \varepsilon$ and only $O(\bar g^1)$ kept:

$$Z_{\bar g} \;=\; (1 + 5\bar g) - 4\cdot\tfrac{1}{2}\bar g - 2\cdot(-\tfrac{1}{2}\bar g) + O(\bar g^2) \;=\; 1 + (5 - 2 + 1)\,\bar g \;=\; 1 + 6\bar g.$$

The β-function follows from $\bar g_0 = \mu^\varepsilon\, \bar g\, Z_{\bar g}$ with $\mu\partial_\mu \bar g_0 = 0$:

$$\boxed{\beta_{\bar g} \;=\; -\varepsilon\bar g \;+\; 6\bar g^2, \qquad \bar g^* \;=\; \tfrac{\varepsilon}{6}.}$$

The anomalous dimensions $\gamma_X = \mu\partial_\mu \ln Z_X$:

$$\gamma_\phi = \tfrac{1}{2}\bar g^* = \tfrac{\varepsilon}{12}, \quad \gamma_D = -\tfrac{1}{2}\bar g^* = -\tfrac{\varepsilon}{12}, \quad \gamma_r = \bar g^* = \tfrac{\varepsilon}{6},$$

and the standard relations

$$\frac{1}{\nu} = 2 - (\gamma_r - \gamma_D) = 2 - \tfrac{\varepsilon}{4}, \quad z = 2 + \gamma_D = 2 - \tfrac{\varepsilon}{12}, \quad \eta = -2\gamma_\phi - \gamma_D = -\tfrac{\varepsilon}{12}.$$

Match. □

Honest scoreboard

The 8-step ledger, audited:

#	Step	What goes *in*	What CFAC produces
1	Diagram inventory	$\phi(G) = 1 + G^2$ from action	Lagrange counts $\{1, 2\}$: bubble + triangle
2	Bubble symmetry factor	$\lvert\mathrm{Aut}(B)\rvert = 2$ from graph	$s(B) = 1/2$ via ϕ-tree theorem
3	ω-int. + spatial routing	causal Reggeon propagator	3a: `OmegaIntegration` ($\Psi' = 2\alpha_0$); 3b: Symanzik routing (X)
4	**Master $\mathcal{I}_{-O-}$**	**Schwinger integral**	— (this is the *one* integration)
5	Triangle $= -\frac{1}{2}\partial_r$ bubble	$\partial_r(p^2 + r)^{-1}$ identity	**the IBP relation itself**
6	Define $\bar{g}$	dim-reg, ε-expansion	rescaling
7	Four Z-projections	Z-system from RG framework	$\{2, 1, \frac{1}{2}, 4\}$ from $\partial_X \mathcal{I}$
8	Solve, β, exponents	Reggeon Ward + η, z, ν dictionary	$\bar{g}^*, \gamma_X, \eta, z, \nu$

Counts (out of 8):

- **1 step is integration** (step 4): one Schwinger / Feynman-parameter calculation, value of $\mathcal{I}_{-O-}$. No CFAC content here.
- **1 step is RG framework** (steps 7–8 setup): Z-system structure, anomalous dimensions, exponent dictionary. Standard textbook RG; CFAC consumes it.
- **6 steps are CFAC outputs** (steps 1, 2, 3, 5, 6, 7): diagram inventory via Lagrange, symmetry factor via $\lvert\mathrm{Aut}\rvert$, **ω-integration via `OmegaIntegration` + spatial routing via Symanzik polynomial** (step 3 — the causal-to-Euclidean bridge), the IBP relation, the rescaling, the four projections.

The non-equilibrium $\leftrightarrow$ equilibrium claim. Step 3a, the `OmegaIntegration`, is the entire content of "DP is a non-equilibrium theory" in this calculation. After step 3a, the integrand is Euclidean and indistinguishable in form from a static φ^4 parametric integrand. *Every CFAC tool developed for equilibrium QFT is therefore portable to causal field theories through this one bridge.* This is the substantive claim worth checking at 2 loops and beyond.

The one-line claim (replacing the old "exponents from analytic combinatorics"):

> **One Schwinger integral, one RG framework, one IBP relation + Lagrange counts ⇒ four DP exponents.**
>
> CFAC's contribution is the diagrams, the symmetry factors, the IBP relation, and the master projections. CFAC does not compute the integral; CFAC does not provide the RG framework. It is the production line that connects them.

Hermite-basis aside

The aside (*can AC characterise the wild forest of φ^4 Doi–Peliti diagrams in a Hermite basis?*) is the right next target. Concept sketch in the companion file `hermite_basis.tex`: one marker ξ_n per Hermite mode, vertex polynomial becomes a tensor $V_{n_1 n_2 n_3 n_4}$ with parity selection $\sum n_i \in 2\mathbb{Z}$, multivariate Lagrange iterates the system, sector projections deliver Z-factors. Worked $N \leq 2$ truncation included.

A Why the topological collapse works: Schwinger-substitution proof

This appendix proves that the ω'-integration in step 3 is equivalent to the graph-collapse picture — i.e., that $\Psi'(\alpha_0) = 2\alpha_0$ and $\Phi'(\alpha_0) = 2\alpha_0^2(m_0^2 + m_1^2)$ fall out by graph operation. The argument is short. The point is to verify that the picture in step 3 is correct; once this appendix is read, the picture stands on its own and no integration is needed to extract the reduced graph in any subsequent calculation.

A.1 Schwinger-parametrise both propagators

Let $P_1 = -i(\omega+\omega') + D(k+k')^2 + r$ and $P_2 = i\omega' + Dk'^2 + r$ be the two causal propagators of the bubble. Using $1/P = \int_0^\infty d\alpha\, e^{-\alpha P}$ (valid for $\Re P > 0$):

$$\begin{aligned}\frac{1}{P_1} &= \int_0^\infty d\alpha_1\, e^{-\alpha_1\left[-i(\omega+\omega')+D(k+k')^2+r\right]},\\ \frac{1}{P_2} &= \int_0^\infty d\alpha_2\, e^{-\alpha_2\left[i\omega'+Dk'^2+r\right]}.\end{aligned}$$

A.2 Collect the ω'-dependent phase

The product $P_1^{-1}P_2^{-1}$ carries an ω'-dependent phase $e^{i\omega'(\alpha_1-\alpha_2)}$ (from $e^{+i\alpha_1\omega'}$ in P_1 and $e^{-i\alpha_2\omega'}$ in P_2).

A.3 Integrate ω' against this phase

The remaining factors are ω'-independent, so

$$\int \frac{d\omega'}{2\pi}\, e^{i\omega'(\alpha_1-\alpha_2)} \;=\; \delta(\alpha_1-\alpha_2).$$

This is where the ω-integration becomes a graph operation: the δ forces $\alpha_1 = \alpha_2$, identifying the two Schwinger parameters of the parallel propagators into a single α_0.

A.4 Read off the reduced Symanzik polynomials

After the δ-substitution the integrand is

$$e^{-\alpha_0\left[-i\omega+D(k+k')^2+Dk'^2+2r\right]},$$

which is the parametric integrand of a 1-loop graph with one Schwinger parameter α_0 and Symanzik polynomials

$$\Psi'(\alpha_0) \;=\; 2\alpha_0, \qquad \Phi'(\alpha_0) \;=\; 2\alpha_0^2(m_0^2+m_1^2),$$

where $m_0^2 = D(k+k')^2 + r$ and $m_1^2 = Dk'^2 + r$ are the two original edge masses (kept distinct in Φ'). *This is the OmegaIntegration output the verify-script prints; the graph-collapse picture in step 3 is its visualisation.*

A.5 Why this is graph-topological, not analytic

The collapse $\alpha_1 = \alpha_2 = \alpha_0$ is a *rule on the graph*, not an analytic computation: it says the two parallel edges of the bubble share a single Schwinger parameter. The factor 2 in Ψ' counts the number of edges that got fused; the $m_0^2 + m_1^2$ in Φ' records the inherited mass multi-set. Both are incidence numbers of the original bubble, not values of any integral.

For a 1-loop Reggeon graph with E parallel edges, the same construction gives a single δ per loop, identifying all E edge parameters into one α_0 with $\Psi' = E\alpha_0$ and Φ' a sum over the E edge masses. For multi-loop Reggeon graphs the collapse rule is encoded by the edge-basis matrix E_f of thesis Eq. 2.28.

A.6 The annotated-tadpole vs literal-tadpole distinction

A natural reading of "edge contraction collapses the bubble to a tadpole" would predict $\Psi = \alpha$, $\Phi = \alpha^2 m^2$ — a literal tadpole. The verification script checks this and reports

```
View (B) Topological-contraction (literal tadpole graph):
  Psi  = alpha0
  Phi  = alpha0**2*m_0**2
  Psi_A == Psi_B ?  False
  Phi_A == Phi_B ?  False
```

The literal tadpole is wrong because it loses the second mass and the edge-multiplicity. The correct picture is an *annotated* tadpole: a doubled edge (multiplicity 2 in Ψ') carrying the mass multi-set $\{m_0, m_1\}$ in Φ'. *Step 3 in the main text uses this annotated picture; the appendix exists to verify that the annotation is correct.*

A.7 Multi-loop generalisation (sketch)

For an L-loop Reggeon graph with edges grouped into L parallel-edge bundles $\{B_1, \ldots, B_L\}$ (one bundle per loop), the ω'-integration produces L δ-functions identifying the Schwinger parameters within each bundle. The reduced Symanzik polynomials are

$$\Psi'(\alpha_{0,1}, \ldots, \alpha_{0,L}) \;=\; \prod_{\ell} |B_\ell|\, \alpha_{0,\ell}, \qquad \Phi'(\alpha_{0,1}, \ldots, \alpha_{0,L}) \;=\; \sum_{\ell} \Big(|B_\ell|\, \alpha_{0,\ell}^2 \sum_{e \in B_\ell} m_e^2 \Big) + \text{(cross terms)},$$

where $|B_\ell|$ is the number of edges in bundle ℓ. The 2-loop sunset has $L = 1$, $|B| = 3$, giving $\Psi' = 3\alpha_0^2$ (one of the views; the other has $\Psi' = 3\alpha_0$ depending on routing convention — see thesis §2.5.3.2 for the canonical form). Verified in `rdft/ac/gribov/test_causal_reduction.py`.

Punchline of Appendix A. The ω-integration in any 1-loop causal Reggeon graph reduces to one rule: *within each parallel-edge bundle, identify the Schwinger parameters and sum the masses in Φ' with the edge-multiplicity in Ψ'.* This rule is purely combinatorial-topological. CFAC executes it via `OmegaIntegration`; the integration of one frequency per loop is replaced by the edge-identification rule on the graph. *Combinatorics and topology have avoided one integration per loop, exactly as AC promises.*